

SAMADH HIGHER SECONDARY SCHOOL
KHAJANAGAR, TRICHY-20
QUARTERLY EXAMINATIONS (2026-27)

CLASS: XII

MARKS: 80

SUBJECT: MATHEMATICS

TIME: 3Hrs

SECTION – A

[20x1=20]

1. The number of symmetric relation on set $A = \{1,2,3,4\}$ is
a) 2^{10} b) 2^6 c) 2^{12} d) 2^{16}
2. A function $f: R \rightarrow R$ defined as $f(x) = x^2 - 4x + 5$ is
a) injective but not surjective b) surjective but not injective
c) both injective and surjective d) neither injective nor surjective
3. The domain of $\sin^{-1} 2x$ is
a) $[0,1]$ b) $[-1,1]$ c) $\left[-\frac{1}{2}, \frac{1}{2}\right]$ d) $[-2,2]$
4. The value of $\sin^{-1}\left(\cos \frac{33\pi}{5}\right)$ is
a) $\frac{3\pi}{5}$ b) $-\frac{7\pi}{5}$ c) $\frac{\pi}{10}$ d) $-\frac{\pi}{10}$
5. The matrix $P = \begin{bmatrix} 0 & 0 & 4 \\ 0 & 4 & 0 \\ 4 & 0 & 0 \end{bmatrix}$ is a
a) square matrix b) diagonal matrix
c) unit matrix d) scalar matrix
6. If $A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$, then A^2 is equal to
a) $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ b) $\begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix}$ c) $\begin{bmatrix} 0 & 1 \\ 0 & 1 \end{bmatrix}$ d) $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$
7. If $[x \ 2 \ 0] \begin{bmatrix} 5 \\ -1 \\ x \end{bmatrix} = [3 \ 1] \begin{bmatrix} -2 \\ x \end{bmatrix}$, then the value of x is
a) -1 b) 0 c) 1 d) 2
8. If $\begin{vmatrix} 2x & 5 \\ 8 & x \end{vmatrix} = \begin{vmatrix} 6 & -2 \\ 7 & 3 \end{vmatrix}$, the value of x is/are
a) 3 b) ± 3 c) ± 6 d) 6
9. If A and B are square matrices of order 3×3 such that $|A| = 5$ and $|B| = 3$,
then $|3AB| =$
a) 135 b) 45 c) 405 d) none of these

10. If A is a square matrix of order 2 and $|A| = -2$ then the value of $|5A^T|$ is

- a) -50 b) -10 c) 10 d) 50

11. If $f(x) = \begin{cases} x \sin 1/x, & x \neq 0 \\ k, & x = 0 \end{cases}$ is continuous at $x = 0$, then the value of k is

- a) 1 b) -1 c) 0 d) 2

12. If $x^y = e^{x-y}$, then dy/dx is

- a) $\frac{1+x}{1+\log x}$ b) $\frac{1-\log x}{1+\log x}$ c) not defined d) $\log x/(1 + \log x)^2$

13. If $y = \sqrt{\sin x + y}$, then $\frac{dy}{dx} =$

- a) $\frac{\sin x}{2y-1}$ b) $\frac{\sin x}{1-2y}$ c) $\frac{\cos x}{1-2y}$ d) $\frac{\cos x}{2y-1}$

14. If $x = A \cos 4t + B \sin 4t$, then $\frac{d^2x}{dt^2}$ is equal to

- a) x b) -x c) 16x d) -16x

15. If the rate of change of area of a circle is equal to the rate of change of its diameter, then its radius is equal to

- a) $2/\pi$ unit b) $1/\pi$ unit c) $\pi/2$ unit d) π units

16. The function $f(x) = x^3 + 3x$ is increasing in the interval.

- a) $(-\infty, 0)$ b) $(0, \infty)$ c) R d) (0,1)

17. Given set $A = \{1,2,3\}$ and a relation $R = \{(1,2), (2,1)\}$, the relation R will be

- a) reflexive if (1,1) is added b) symmetric if (2,3) is added
c) transitive if (1,1) is added d) symmetric if (3,2) is added

18. $\tan^{-1} \left\{ \sin \left(-\frac{\pi}{2} \right) \right\}$ is equal to

- a) -1 b) 1 c) $\frac{\pi}{2}$ d) $-\frac{\pi}{4}$

Assertion and Reason Questions:

- a) Both A and R are true and R is the correct explanation of A.
b) Both A and R are true and R is not the correct explanation of A.
c) A is true but R is false.
d) A is false but R is true.

Assertion (A) : For matrix $A = [a_{ij}]_{2 \times 3}$ where $a_{ij} = 3j - 5i$, $a_{12} = -7$

Reason (R) : $a_{12} = -1$ as $i = 1, j = 2$

20. **Assertion (A) :** Zero matrices $\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ and $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$ are not equal.

Reason (R) : Two matrices are said to be equal if their orders are same and corresponding elements are equal.

SECTION - B

[5x2=10]

21. If $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 3 & 1 \\ 2 & 5 \end{bmatrix} = \begin{bmatrix} 7 & 11 \\ k & 23 \end{bmatrix}$, then write the value of k .

22. i) Write A^{-1} for $A = \begin{bmatrix} 2 & 5 \\ 1 & 3 \end{bmatrix}$

(OR)

ii) Find the value of x such that the points $(0,2)$, $(1,x)$ and $(3,1)$ are collinear.

23. Determine the value of the constant 'k' so that the function.

$$f(x) = \begin{cases} \frac{kx}{|x|}, & \text{if } x < 0 \text{ is continuous} \\ 3, & \text{if } x \geq 0 \end{cases} \quad \text{at } x = 0.$$

24. Find $\frac{dy}{dx}$, if $\sin y + x = \log x$.

25. If the rate of change of volume of a sphere is equal to the rate of change of its radius, then find the radius.

SECTION - C

[6x3=18]

26. Show that the relation R in the set of real numbers, defined as $R = \{(a,b) : a \leq b^2\}$ is neither reflexive nor symmetric nor transitive.

27. Write the value of $\tan^{-1} \left[2 \sin \left(2 \cos^{-1} \frac{\sqrt{3}}{2} \right) \right]$.

28. Write the principal value of $\sin^{-1} \left[\sin \left(-\frac{17\pi}{8} \right) \right]$.

29. i) Find the matrix X , for which $\begin{bmatrix} 5 & 4 \\ 1 & 1 \end{bmatrix} X = \begin{bmatrix} 1 & -2 \\ 1 & 3 \end{bmatrix}$.

(OR)

ii) For the matrices A and B , verify that $(AB)' = B'A'$ where $A = \begin{bmatrix} 1 \\ -4 \\ 3 \end{bmatrix}$, $B = \begin{bmatrix} -1 & 2 & 1 \end{bmatrix}$

30. If $A = \begin{bmatrix} 2 & 3 \\ 5 & -2 \end{bmatrix}$ be such that $A^{-1} = kA$, then find the value of k .

31. Find $\frac{dy}{dx}$, if $y = (\sin x)^{\tan x} + (\cos x)^{\sec x}$.

SECTION – D

[4x5=20]

32. i) If $A = \begin{bmatrix} 1 & 2 & -3 \\ 2 & 0 & -3 \\ 1 & 2 & 0 \end{bmatrix}$, then find A^{-1} and hence solve the following system of equations

$$x + 2y - 3z = 1, \quad 2x - 3z = 2, \quad x + 2y = 3.$$

(OR)

- ii) Find the product of the matrices $\begin{bmatrix} 1 & 2 & -3 \\ 2 & 3 & 2 \\ 3 & -3 & -4 \end{bmatrix} \begin{bmatrix} -6 & 17 & 13 \\ 14 & 5 & -8 \\ -15 & 9 & -1 \end{bmatrix}$ and hence solve the system of linear equations.

$$x + 2y - 3z = 4, \quad 2x + 3y + 2z = 2, \quad 3x - 3y - 4z = 11.$$

33. If $x = a \sin^3 \theta$, $y = b \cos^3 \theta$, then find $\frac{d^2y}{dx^2}$ at $\theta = \frac{\pi}{4}$.

34. Let $A = R - \{5\}$ and $B = R - \{1\}$. Consider the function $f: A \rightarrow B$, defined by $f(x) = \frac{x-3}{x-5}$. Show that f is one-one and onto.

35. i) Find $\frac{dy}{dx}$, $(\cos x)^y = (\sin y)^x$

- ii) If $y = Ae^{mx} + Be^{nx}$, prove that $\frac{d^2y}{dx^2} - (m+n)\frac{dy}{dx} + mny = 0$.

SECTION – E

CASE STUDY

[3x4=12]

36. In a city there are two factories A and B. Each factory produces sports clothes for boys and girls. There are three types of clothes produced in both the factories type I, type II and type III. For boys the number of units of type I, II and III respectively are 80, 70 and 65 in factory A and 85, 65 and 72 are in factory B. For girls the number of units of types I, II and III respectively are 80, 75, 90 in factory A and 50, 55, 80 are in factory B.

- i) Write the matrix P, if P represents the matrix of number of units of each type produced by factory A for both boys and girls.
- ii) Write the matrix Q, if Q represents the matrix of number of units of each type produced by factory B for both boys and girls.
- iii) Find the total production of sports clothes of each type for boys.

Gautam buys 5 pens, 3 bags and 1 instrument box and pays a sum of ₹160. From the same shop. Vikram buys 2 pens, 1 bag and 3 instrument boxes and pays a sum of ₹190. Also Ankur buys 1 pen, 2 bags and 4 instrument boxes and pays a sum of ₹250.

	Pen	Bags	Instrument
Gautam	5	3	1
Vikram	2	1	3
Ankur	1	2	4

i) Convert the given above situation into a matrix equation of the form $AX = B \rightarrow (1)$

ii) Find $|A| \rightarrow (1)$

iii) Find $A^{-1} \rightarrow (2)$

38. Ms. Remka of city school is teaching chain rule to her students with the help of flow-chart.

$$f(x) = g(h(x)), \text{ then}$$

$$f'(x) = [g(h(x))]' h'(x)$$

i) Find $\frac{d}{dx}(\cos x^5) \rightarrow (1)$

ii) Find $\frac{d}{dx} \sin(\cos x) \rightarrow (1)$

iii) Find $\frac{d}{dx}(\sin 2x)$ at $x = \frac{\pi}{2} \rightarrow (2)$